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Table of Contents

Message from the ICSE 2023 General Chair ..... **xxxvi**
Message from the ICSE 2023 Program Co-Chairs ..... **xl**
Technical Track Program Committee ..... **xliv**
Additional Reviewers ..... **liv**
Workshops Committee ..... **lvi**
Artifact Evaluation Committee ..... **lvii**
Most Influential Paper (MIP) Award ..... **lix**

Keynotes

Future Software for Life in Trusted Futures 1
Sarah Pink (Monash University, Australia)

The Road Toward Dependable AI Based Systems 2
Paolo Tonella (Università della Svizzera italiana, Switzerland)

Software Engineering as the Linchpin of Responsible AI 3
Liming Zhu (CSIRO's Data61, Australia; University of New South Wales, Australia)

AI models for SE

One Adapter for All Programming Languages? Adapter Tuning for Code Search and Summarization 5
Deze Wang (National University of Defense Technology, China), Boxing Chen (n/a), Shanshan Li (National University of Defense Technology, China), Wei Luo (n/a), Shaoliang Peng (Hunan University, China), Wei Dong (National University of Defense Technology, China), and Xiangke Liao (National University of Defense Technology, China)

CCRep: Learning Code Change Representations via Pre-Trained Code Model and Query Back 17
Zhongxin Liu (Zhejiang University, China), Zhijie Tang (Zhejiang Univeristy, China), Xin Xia (Huawei, China), and Xiaohu Yang (Zhejiang University, China)

Keeping Pace with Ever-Increasing Data: Towards Continual Learning of Code Intelligence Models 30
Shuzheng Gao (Harbin institute of technology, China), Hongyu Zhang (Chongqing University, China), Cuiyun Gao (Harbin Institute of Technology, China), and Chaozheng Wang (Harbin Institute of Technology, China)

Fuzzing: Applications

Detecting JVM JIT Compiler Bugs via Exploring Two-Dimensional Input Spaces 43
Haoxiang Jia (Huazhong University of Science and Technology, China), Ming Wen (Huazhong University of Science and Technology, China), Zifan Xie (Huazhong University of Science and Technology, China), Xiaochen Guo (Huazhong University of Science and Technology, China), Rongxin Wu (Xiamen University, China), Maolin Sun (Huazhong University of Science and Technology, China), Kang Chen (Huazhong University of Science and Technology, China), and Hai Jin (Huazhong University of Science and Technology, China)

JITfuzz: Coverage-Guided Fuzzing for JVM Just-in-Time Compilers 56
Mingyuan Wu (Southern University of Science and Technology and the University of Hong Kong, China), Minghai Lu (Southern University of Science and Technology, China), Heming Cui (The University of Hong Kong, China), Junjie Chen (Tianjin University, China), Yuqun Zhang (Southern University of Science and Technology, China), and Lingming Zhang (University of Illinois Urbana-Champaign, USA)

Validating SMT Solvers via Skeleton Enumeration Empowered by Historical Bug-Triggering Inputs 69
Maolin Sun (Huazhong University of Science and Technology, China), Yibiao Yang (Nanjing University, China), Ming Wen (Huazhong University of Science and Technology, China), Yongcong Wang (Huazhong University of Science and Technology, China), Yuming Zhou (Nanjing University, China), and Hai Jin (Huazhong University of Science and Technology, China)

Regression Fuzzing for Deep Learning Systems 82
Hanmo You (Tianjin University, China), Zan Wang (Tianjin University, China), Junjie Chen (Tianjin University, China), Shuang Liu (Tianjin University, China), and Shuochuan Li (Tianjin University, China)

Operand-Variation-Oriented Differential Analysis for Fuzzing Binding Calls in PDF Readers 95
Suyue Guo (Renmin University of China, China), Xinyu Wan (Renmin University of China, China), Wei You (Renmin University of China, China), Bin Liang (Renmin University of China, China), Wenchang Shi (Renmin University of China, China), Yiwei Zhang (Renmin University of China, China), Jianjun Huang (Renmin University of China, China), and Jian Zhang (State Key Laboratory of Computer Science, Institute of Software, Chinese Academy of Sciences, China)

Mining software repositories

The Untold Story of Code Refactoring Customizations in Practice 108
Daniel Oliveira (Pontifical Catholic University of Rio de Janeiro, Brazil), Wesley K. G. Assunção (Johannes Kepler University, Austria; Pontifical Catholic University of Rio de Janeiro, Brazil), Alessandro Garcia (Pontifical Catholic University of Rio de Janeiro, Brazil), Ana Carla Bibiano (Pontifical Catholic University of Rio de Janeiro, Brazil), Márcio Ribeiro (Federal University of Alagoas, Brazil), Rohit Gheyi (Federal University of Campina Grande, Brazil), and Baldoino Fonseca (Federal University of Alagoas, Brazil)

Data Quality for Software Vulnerability Datasets 121
Roland Croft (The University of Adelaide, Australia; Cyber Security Cooperative Research Centre, Australia), M. Ali Babar (The University of Adelaide, Australia; Cyber Security Cooperative Research Centre, Australia), and Mehdi Kholoosi (The University of Adelaide, Australia; Cyber Security Cooperative Research Centre, Australia)

Do Code Refactorings Influence the Merge Effort? 134
André Oliveira (Universidade Federal Fluminense, Brazil), Vânia Neves (Universidade Federal Fluminense, Brazil), Alexandre Plastino (Universidade Federal Fluminense, Brazil), Ana Carla Bibiano (PUC-Rio, Brazil), Alessandro Garcia (PUC-Rio, Brazil), and Leonardo Murta (Universidade Federal Fluminense, Brazil)

A Comprehensive Study of Real-World Bugs in Machine Learning Model Optimization 147
Hao Guan (The University of Queensland, Australia; Southern University of Science and Technology, China), Ying Xiao (Southern University of Science and Technology, China), Jiaying Li (Microsoft Software Technology Center Asia, China), Yepang Liu (Southern University of Science and Technology, China), and Guangdong Bai (The University of Queensland, Australia)

Fault localization

Evaluating the Impact of Experimental Assumptions in Automated Fault Localization 159
Ezekiel Soremekun (Royal Holloway, University of London, UK; SnT), Lukas Kirschner (Saarland University, Germany), Marcel Böhme (MPI-SP, Germany), and Mike Papadakis (SnT, University of Luxembourg, Luxembourg)

Locating Framework-Specific Crashing Faults with Compact and Explainable Candidate Set 172
Jiwei Yan (Institute of Software, Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Miaomiao Wang (Institute of Software, Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Yepang Liu (Southern University of Science and Technology, China), Jun Yan (Institute of Software, Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), and Long Zhang (Institute of Software, Chinese Academy of Sciences, China)

PExReport: Automatic Creation of Pruned Executable Cross-Project Failure Reports 184
Sunzhou Huang (University of Texas at San Antonio, USA) and Xiaoyin Wang (University of Texas at San Antonio, USA)

RAT: A Refactoring-Aware Traceability Model for Bug Localization 196
Feifei Niu (Nanjing University, China), Wesley K. G. Assunção (Johannes Kepler University, Austria), Liguo Huang (Southern Methodist University, USA), Christoph Mayr-Dorn (Johannes Kepler University, Austria), Jidong Ge (Nanjing University, China), Bin Luo (Nanjing University, China), and Alexander Egyed (Johannes Kepler University, Austria)

Formal verification

How Do We Read Formal Claims? Eye-Tracking and the Cognition of Proofs About Algorithms ... 208
Hammad Ahmad (University of Michigan, USA), Zachary Karas (University of Michigan, USA), Kimberly Diaz (University of Michigan, USA), Amir Kamil (University of Michigan, USA), Jean-Baptiste Jeannin (University of Michigan, USA), and Westley Weimer (University of Michigan, USA)

Which of My Assumptions are Unnecessary for Realizability and Why Should I Care? 221
Rafi Shalom (Tel Aviv University, Israel) and Shahar Maoz (Tel Aviv University, Israel)

APIs and libraries

UpCy: Safely Updating Outdated Dependencies 233
Andreas Dann (CodeShield GmbH, Germany), Ben Hermann (Technical University Dortmund, Germany), and Eric Bodden (Heinz Nixdorf Institute & Fraunhofer IEM, Germany)

APICAD: Augmenting API Misuse Detection through Specifications from Code and Documents ... 245
Xiaoke Wang (Wuhan University, China) and Lei Zhao (Wuhan University, China)

Compatibility Issue Detection for Android Apps Based on Path-Sensitive Semantic Analysis 257
Sen Yang (Army Engineering University of PLA, China), Sen Chen (Tianjin University, China), Lingling Fan (Nankai University, China), Sihan Xu (Nankai University, China), Zhanwei Hui (Academy of Military Science, China), and Song Huang (Army Engineering University of PLA, China)

OSSFP: Precise and Scalable C/C++ Third-Party Library Detection Using Fingerprinting Functions 270
Jiahui Wu (Nanyang Technological University, Singapore), Zhengzi Xu (Nanyang Technological University, Singapore), Wei Tang (Tsinghua University, China), Lyuye Zhang (Nanyang Technological University, Singapore), Yueming Wu (Nanyang Technological University, Singapore), Chengyue Liu (Nanyang Technological University, Singapore), Kairan Sun (Nanyang Technological University, Singapore), Lida Zhao (Nanyang Technological University, Singapore), and Yang Liu (Nanyang Technological University, Singapore)

Blockchain/Smart contracts

SmartMark: Software Watermarking Scheme for Smart Contracts 283
Taeyoung Kim (Sungkyunkwan University, South Korea), Yunhee Jang (Sungkyunkwan University, South Korea), Chanjong Lee (Sungkyunkwan University, South Korea), Hyungjoon Koo (Sungkyunkwan University, South Korea), and Hyoungshick Kim (Sungkyunkwan University, South Korea)

Turn the Rudder: A Beacon of Reentrancy Detection for Smart Contracts on Ethereum 295
Zibin Zheng (School of Software Engineering, Sun Yat-sen University, China), Neng Zhang (School of Software Engineering, Sun Yat-sen University, China), Jianzhong Su (School of Software Engineering, Sun Yat-sen University, China), Zhijie Zhong (School of Software Engineering, Sun Yat-sen University, China), Mingxi Ye (School of Software Engineering, Sun Yat-sen University, China), and Jiachi Chen (School of Software Engineering, Sun Yat-sen University, China)

BSHUNTER: Detecting and Tracing Defects of Bitcoin Scripts 307
Peilin Zheng (Sun Yat-sen University), Xiapu Luo (The Hong Kong Polytechnic University), and Zibin Zheng (Sun Yat-sen University)

Cognitive aspects of software development

Do I Belong? Modeling Sense of Virtual Community Among Linux Kernel Contributors 319
Bianca Trinkenreich (Northern Arizona University, USA), Klaas-Jan Stol (University College Cork, Ireland), Anita Sarma (Oregon State University, USA), Daniel M. German (University of Victoria, Canada), Marco A. Gerosa (Northern Arizona University, USA), and Igor Steinmacher (Northern Arizona University, USA)

Code smells and clones

Comparison and Evaluation of Clone Detection Techniques with Different Code Representations 332
Yuekun Wang (University of Science and Technology of China, China), Yuhang Ye (University of Science and Technology of China, China), Yueming Wu (Nanyang Technological University, Singapore), Weiwei Zhang (University of Science and Technology of China, China), Yinxing Xue (University of Science and Technology of China, China), and Yang Liu (Nanyang Technological University, Singapore)

Learning Graph-Based Code Representations for Source-Level Functional Similarity Detection..... 345
Jiahao Liu (National University of Singapore, Singapore), Jun Zeng (National University of Singapore, Singapore), Xiang Wang (University of Science and Technology of China, China), and Zhenkai Liang (National University of Singapore, Singapore)

The Smelly Eight: An Empirical Study on the Prevalence of Code Smells in Quantum Computing.. 358
Qihong Chen (University of California, USA), Rúben Câmara (LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal), José Campos (Faculty of Engineering of University of Porto, Portugal; LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal), André Souto (LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal; Instituto de Telecomunicações, Portugal), and Iftekhar Ahmed (University of California, USA)

Fuzzing: Techniques and tools

Reachable Coverage: Estimating Saturation in Fuzzing 371
Danushka Liyanage (Monash University, Australia), Marcel Böhme (Max Planck Institute for Security and Privacy (MPI-SP), Germany and Monash University, Australia), Chakkrit Tantithamthavorn (Monash University), and Stephan Lipp (TU Munich, Germany)

Learning Seed-Adaptive Mutation Strategies for Greybox Fuzzing 384
Myungho Lee (Korea University, South Korea), Sooyoung Cha (Sungkyunkwan University, South Korea), and Hakjoo Oh (Korea University, South Korea)

Improving Java Deserialization Gadget Chain Mining via Overriding-Guided Object Generation... 397
Sicong Cao (Yangzhou University, China), Xiaobing Sun (Yangzhou University, China), Xiaoxue Wu (Yangzhou University, China), Lili Bo (Yangzhou University, China), Bin Li (Yangzhou University, China), Rongxin Wu (Xiamen University, China), Wei Liu (Yangzhou University, China), Biao He (Ant Group, China), Yu Ouyang (Ant Group, China), and Jiajia Li (Ant Group, China)

Evaluating and Improving Hybrid Fuzzing 410
Ling Jiang (Southern University of Science and Technology, China), Hengchen Yuan (Southern University of Science and Technology, China), Mingyuan Wu (Southern University of Science and Technology, China), Lingming Zhang (University of Illinois Urbana-Champaign, USA), and Yuqun Zhang (Southern University of Science and Technology, China)

Software architectures and design

Robustification of Behavioral Designs Against Environmental Deviations 423
Changjian Zhang (Carnegie Mellon University, USA), Tarang Saluja (Swarthmore College, USA), Rômulo Meira-Góes (The Pennsylvania State University, USA), Matthew Bolton (University of Virginia, USA), David Garlan (Carnegie Mellon University, USA), and Eunsuk Kang (Carnegie Mellon University, USA)

A Qualitative Study on the Implementation Design Decisions of Developers 435
Jenny T. Liang (Carnegie Mellon University, USA), Maryam Arab (George Mason University, USA), Minhyuk Ko (Virginia Tech, USA), Amy J. Ko (University of Washington, USA), and Thomas D. LaToza (George Mason University, USA)

Software security and privacy

BFTDetector: Automatic Detection of Business Flow Tampering for Digital Content Service 448
I Luk Kim (Purdue University, USA), Weihang Wang (University of Southern California, USA), Yonghwi Kwon (University of Virginia, USA), and Xiangyu Zhang (Purdue University, USA)

FedSlice: Protecting Federated Learning Models from Malicious Participants with Model Slicing 460
Ziqi Zhang (Peking University, China), Yuanchun Li (Tsinghua University, China), Bingyan Liu (Beijing University of Posts and Telecommunications, China), Yifeng Cai (Peking University, China), Ding Li (Peking University, China), Yao Guo (Peking University, China), and Xiangqun Chen (Peking University, China)

PTPDroid: Detecting Violated User Privacy Disclosures to Third-Parties of Android Apps 473
Zeya Tan (Nanjing University of Science and Technology, China) and Wei Song (Nanjing University of Science and Technology, China)

AdHere: Automated Detection and Repair of Intrusive Ads 486
Yutian Yan (University of Southern California, USA), Yunhui Zheng (IBM T. J. Watson Research Center, USA), Xinyue Liu (University at Buffalo SUNY, USA), Nenad Medvidovic (University of Southern California, USA), and Weihang Wang (University of Southern California, USA)

Bad Snakes: Understanding and Improving Python Package Index Malware Scanning 499
Duc-Ly Vu (Chainguard and FPT University), Zachary Newman (Chainguard), and John Speed Meyers (Chainguard)

AI systems engineering

FedDebug: Systematic Debugging for Federated Learning Applications 512
Waris Gill (Virginia Tech, USA), Ali Anwar (University of Minnesota Twin Cities, USA), and Muhammad Ali Gulzar (Virginia Tech, USA)

Practical and Efficient Model Extraction of Sentiment Analysis APIs 524
Weibin Wu (Sun Yat-sen University, China), Jianping Zhang (The Chinese University of Hong Kong, China), Victor Junqiu Wei (The Hong Kong Polytechnic University, China), Xixian Chen (Tencent, China), Zibin Zheng (Sun Yat-sen University, China), Irwin King (The Chinese University of Hong Kong, China), and Michael R. Lyu (The Chinese University of Hong Kong, China)

CrossCodeBench: Benchmarking Cross-Task Generalization of Source Code Models 537
Changan Niu (Nanjing University, China), Chuanyi Li (Nanjing University, China), Vincent Ng (University of Texas at Dalla, USA), and Bin Luo (Nanjing University, China)

Debugging

ECSTATIC: An Extensible Framework for Testing and Debugging Configurable Static Analysis 550
Austin Mordahl (The University of Texas at Dallas, USA), Zenong Zhang (The University of Texas at Dallas, USA), Dakota Soles (The University of Texas at Dallas, USA), and Shiyi Wei (The University of Texas at Dallas, USA)

Responsibility in Context: On Applicability of Slicing in Semantic Regression Analysis 563
Sahar Badihi (University of British Columbia, Canada), Khaled Ahmed (University of British Columbia, Canada), Yi Li (Nanyang Technological University, Singapore), and Julia Rubin (University of British Columbia, Canada)

Does the Stream API Benefit from Special Debugging Facilities? A Controlled Experiment on Loops and Streams with Specific Debuggers 576
Jan Reichl (University of Duisburg-Essen, Germany), Stefan Hanenberg (University of Duisburg-Essen, Germany), and Volker Gruhn (University of Duisburg-Essen, Germany)

Fonte: Finding Bug Inducing Commits from Failures 589
Gabin An (KAIST, Republic of Korea), Jingun Hong (SAP Labs Korea, Republic of Korea), Naryeong Kim (KAIST, Republic of Korea), and Shin Yoo (KAIST, Republic of Korea)

Defect analysis

RepresentThemAll: A Universal Learning Representation of Bug Reports 602
Sen Fang (Macau University of Science and Technology, China), Tao Zhang (Macau University of Science and Technology, China), Youshuai Tan (Macau University of Science and Technology, China), He Jiang (Dalian University of Technology, China), Xin Xia (Huawei, China), and Xiaobing Sun (Yangzhou University, China)

Demystifying Exploitable Bugs in Smart Contracts 615
Zhuo Zhang (Purdue University), Brian Zhang (Harrison High School), Wen Xu (Georgia Institute of Technology), and Zhiqiang Lin (Ohio State University)

Understanding and Detecting On-the-Fly Configuration Bugs 628
Teng Wang (National University of Defense Technology, China), Zhouyang Jia (National University of Defense Technology, China; Hunan University, China), Shanshan Li (National University of Defense Technology, China), Si Zheng (National University of Defense Technology, China), Yue Yu (National University of Defense Technology, China), Erci Xu (National University of Defense Technology, China), Shaoliang Peng (Hunan University, China), and Xiangke Liao (National University of Defense Technology, China)

Explaining Software Bugs Leveraging Code Structures in Neural Machine Translation 640
Parvez Mahbub (Dalhousie University, Canada), Ohiduzzaman Shuvo (Dalhousie University, Canada), and Mohammad Masudur Rahman (Dalhousie University, Canada)

Developers' behaviors

Is It Enough to Recommend Tasks to Newcomers? Understanding Mentoring on Good First Issues 653
Xin Tan (Beihang University, China), Yiran Chen (Beihang University, China), Haohua Wu (Beihang University, China), Minghui Zhou (Peking University, China), and Li Zhang (Beihang University, China)

From Organizations to Individuals: Psychoactive Substance Use By Professional Programmers 665
Kaia Newman (University of Michigan, USA), Madeline Endres (University of Michigan, USA), Westley Weimer (University of Michigan, USA), and Brittany Johnson (George Mason University, USA)

On the Self-Governance and Episodic Changes in Apache Incubator Projects: An Empirical Study 678
Likang Yin (University of California, Davis, USA), Xiyu Zhang (University of California, Davis, USA), and Vladimir Filkov (University of California, Davis, USA)

Socio-Technical Anti-Patterns in Building ML-Enabled Software: Insights from Leaders on the Forefront 690
Alina Mailach (Leipzig University, Germany) and Norbert Siegmund (Leipzig University, Germany)

Moving on from the Software Engineers' Gambit: an Approach to Support the Defense of Software Effort Estimates 703
Patricia G.F Matsubara (Federal University of Amazonas (UFAM); Federal University of Mato Grosso do Sul (UFMS), Brazil), Igor Steinmacher (Northern Arizona University, USA), Bruno Gadelha (Federal University of Amazonas (UFAM), Brazil), and Tayana Conte (Federal University of Amazonas (UFAM), Brazil)

Program translation and synthesis

Concrat: An Automatic C-to-Rust Lock API Translator for Concurrent Programs 716
Jaemin Hong (KAIST, South Korea) and Sukyoung Ryu (KAIST, South Korea)

Triggers for Reactive Synthesis Specifications 729
Gal Amram (Tel Aviv University, Israel), Dor Ma'ayan (Tel Aviv University, Israel), Shahar Maoz (Tel Aviv University, Israel), Or Pistiner (Tel Aviv University, Israel), and Jan Oliver Ringert (Bauhaus University Weimar, Germany)

Using Reactive Synthesis: An End-to-End Exploratory Case Study 742
Dor Ma'ayan (Tel Aviv University, Israel) and Shahar Maoz (Tel Aviv University, Israel)

Syntax and Domain Aware Model for Unsupervised Program Translation 755
Fang Liu (Beihang University, China), Jia Li (Peking University, China), and Li Zhang (Beihang University, China)

Documentation

Developer-Intent Driven Code Comment Generation 768
Fangwen Mu (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Xiao Chen (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Lin Shi (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Song Wang (York University, Canada), and Qing Wang (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China)

Data Quality Matters: A Case Study of Obsolete Comment Detection 781
Shengbin Xu (State Key Lab for Novel Software Technology, Nanjing University, China), Yuan Yao (State Key Lab for Novel Software Technology, Nanjing University, China), Feng Xu (State Key Lab for Novel Software Technology, Nanjing University, China), Tianxiao Gu (Tiktok, USA), Jingwei Xu (State Key Lab for Novel Software Technology, Nanjing University, China), and Xiaoxing Ma (State Key Lab for Novel Software Technology, Nanjing University, China)

Revisiting Learning-Based Commit Message Generation 794
Jinhao Dong (Peking University, China), Yiling Lou (Fudan University, China), Dan Hao (Peking University, China), and Lin Tan (Purdue University, USA)

Commit Message Matters: Investigating Impact and Evolution of Commit Message Quality 806
Jiawei Li (University of California, Irvine, USA) and Iftekhar Ahmed (University of California, Irvine, USA)

Software logging

PILAR: Studying and Mitigating the Influence of Configurations on Log Parsing 818
Hetong Dai (Concordia University, Canada), Yiming Tang (Concordia University, Canada), Heng Li (Polytechnique Montréal, Canada), and Weiyi Shang (Concordia University, Canada)

Did We Miss Something Important? Studying and Exploring Variable-Aware Log Abstraction 830
Zhenhao Li (Concordia University, Canada), Chuan Luo (Microsoft Research, China), Tse-Hsun Chen (Concordia University, Canada), Weiyi Shang (Concordia University, Canada), Shilin He (Microsoft Research, China), Qingwei Lin (Microsoft Research, China), and Dongmei Zhang (Microsoft Research, China)

On the Temporal Relations Between Logging and Code 843
Zishuo Ding (Concordia University, Canada), Yiming Tang (Concordia University, Canada), Yang Li (Beijing University of Posts and Telecommunications, China), Heng Li (Polytechnique Montreal, Canada), and Weiyi Shang (Concordia University, Canada)

How Do Developers' Profiles and Experiences Influence their Logging Practices? An Empirical Study of Industrial Practitioners 855
Guoping Rong (Nanjing University), Shenghui Gu (Nanjing University), Haifeng Shen (Australian Catholic University), He Zhang (Nanjing University), and Hongyu Kuang (Nanjing University)

When to Say What: Learning to Find Condition-Message Inconsistencies 868
Islem Bouzenia (University of Stuttgart, Germany) and Michael Pradel (University of Stuttgart, Germany)

SemParser: A Semantic Parser for Log Analytics 881
Yintong Huo (The Chinese University of Hong Kong, China), Yuxin Su (Sun Yat-sen University, China), Cheryl Lee (The Chinese University of Hong Kong, China), and Michael R. Lyu (The Chinese University of Hong Kong, China)

Test generation

BADGE: Prioritizing UI Events with Hierarchical Multi-armed Bandits for Automated UI Testing 894
Dezhi Ran (Peking University, China), Hao Wang (Peking University, China), Wenyu Wang (University of Illinois Urbana-Champaign, USA), and Tao Xie (Peking University, China)

Efficiency Matters: Speeding Up Automated Testing with GUI Rendering Inference 906
Sidong Feng (Monash University, Australia), Mulong Xie (Australian National University, Australia), and Chunyang Chen (Monash University, Australia)

CodaMOSA: Escaping Coverage Plateaus in Test Generation with Pre-Trained Large Language Models 919
Caroline Lemieux (University of British Columbia, Canada), Jeevana Priya Inala (Microsoft Research, USA), Shuvendu K. Lahiri (Microsoft Research, USA), and Siddhartha Sen (Microsoft Research, USA)

SE for security 1

TAINTMINI: Detecting Flow of Sensitive Data in Mini-Programs with Static Taint Analysis 932
Chao Wang (The Ohio State University), Ronny Ko (The Ohio State University), Yue Zhang (The Ohio State University), Yuqing Yang (The Ohio State University), and Zhiqiang Lin (The Ohio State University)

AChecker: Statically Detecting Smart Contract Access Control Vulnerabilities 945
Asem Ghaleb (University of British Columbia, Canada), Julia Rubin (University of British Columbia, Canada), and Karthik Pattabiraman (University of British Columbia, Canada)

Fine-Grained Commit-Level Vulnerability Type Prediction by CWE Tree Structure 957
Shengyi Pan (Zhejiang University, China), Lingfeng Bao (Zhejiang University, China), Xin Xia (Huawei, China), David Lo (Singapore Management University, Singapore), and Shanping Li (Zhejiang University, China)

Silent Vulnerable Dependency Alert Prediction with Vulnerability Key Aspect Explanation 970
Jiamou Sun (CSIRO, Australia), Zhenchang Xing (CSIRO, Australia; Australian National University, Australia), Qinghua Lu (CSIRO, Australia), Xiwei Xu (CSIRO, Australia), Liming Zhu (CSIRO, Australia; University of New South Wales, Australia), Thong Hoang (CSIRO, Australia), and Dehai Zhao (Australian National University, Australia)

Development and evolution of AI-intensive systems

Reusing Deep Neural Network Models Through Model Re-Engineering 983
Binhang Qi (Beihang University, China), Hailong Sun (Beihang University, China), Xiang Gao (Beihang University, China), Hongyu Zhang (Chongqing University, China), Zhaotian Li (Beihang University, China), and Xudong Liu (Beihang University, China)

PYEVOLVE: Automating Frequent Code Changes in Python ML Systems 995
Malinda Dilhara (University of Colorado Boulder, USA), Danny Dig (University of Colorado Boulder, JetBrains Research, USA), and Ameya Ketkar (Uber Technologies Inc., USA)

DeepArc: Modularizing Neural Networks for the Model Maintenance 1008
Xiaoning Ren (University of Science and Technology of China, China), Yun Lin (Shanghai Jiao Tong University, China), Yinxing Xue (University of Science and Technology of China, China), Ruofan Liu (National University of Singapore, Singapore), Jun Sun (Singapore Management University, Singapore), Zhiyong Feng (Tianjin University, China), and Jin Song Dong (National University of Singapore, Singapore)

Decomposing a Recurrent Neural Network into Modules for Enabling Reusability and Replacement 1020
Sayem Mohammad Imtiaz (Iowa State University, USA), Fraol Batole (Iowa State University, USA), Astha Singh (Iowa State University, USA), Rangeet Pan (IBM Research, USA; Iowa State University, USA), Breno Dantas Cruz (Iowa State University, USA), and Hridesh Rajan (Iowa State University, USA)

Vulnerability analysis and assessment

CHRONOS: Time-Aware Zero-Shot Identification of Libraries from Vulnerability Reports 1033
Yunbo Lyu (Singapore Management University, Singapore), Le-Cong Thanh (Singapore Management University, Singapore), Hong Jin Kang (Singapore Management University, Singapore), Ratnadira Widyasari (Singapore Management University, Singapore), Zhipeng Zhao (Singapore Management University, Singapore), Xuan-Bach D. Le (University of Melbourne, Australia), Ming Li (Nanjing University, China), and David Lo (Singapore Management University, Singapore)

Understanding the Threats of Upstream Vulnerabilities to Downstream Projects in the Maven Ecosystem 1046
Yulun Wu (Huazhong University of Science and Technology, China), Zeliang Yu (Huazhong University of Science and Technology, China), Ming Wen (Huazhong University of Science and Technology, China), Qiang Li (Huazhong University of Science and Technology, China), Deqing Zou (Huazhong University of Science and Technology, China), and Hai Jin (Huazhong University of Science and Technology, China)

SecBench.js: An Executable Security Benchmark Suite for Server-Side JavaScript 1059
Masudul Hasan Masud Bhuiyan (CISPA Helmholtz Center for Information Security, Germany), Adithya Srinivas Parthasarathy (Indian Institute of Information Technology, Design and Manufacturing, India), Nikos Vasilakis (Brown University & MIT, United States), Michael Pradel (University of Stuttgart, Germany), and Cristian-Alexandru Staicu (CISPA Helmholtz Center for Information Security, Germany)

On Privacy Weaknesses and Vulnerabilities in Software Systems 1071
Pattaraporn Sangaroonsilp (University of Wollongong, Australia), Hoa Khanh Dam (University of Wollongong, Australia), and Aditya Ghose (University of Wollongong, Australia)

Defect detection and prediction

Detecting Exception Handling Bugs in C++ Programs 1084
Hao Zhang (Institute of Software, Chinese Academy of Sciences, China), Ji Luo (Institute of Software, Chinese Academy of Sciences, China), Mengze Hu (Institute of Software, Chinese Academy of Sciences, China), Jun Yan (Institute of Software, Chinese Academy of Sciences, China), Jian Zhang (Institute of Software, Chinese Academy of Sciences, China), and Zongyan Qiu (Peking University, China)

Learning to Boost Disjunctive Static Bug-Finders 1097
Yoonseok Ko (Meta, United Kingdom) and Hakjoo Oh (Korea University, South Korea)

Predicting Bugs by Monitoring Developers During Task Execution 1110
Gennaro Laudato (University of Molise, Italy), Simone Scalabrino (University of Molise, Italy), Nicole Novielli (University of Bari, Italy), Filippo Lanubile (University of Bari, Italy), and Rocco Oliveto (University of Molise, Italy)

Detecting Isolation Bugs via Transaction Oracle Construction 1123
Wensheng Dou (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences; University of Chinese Academy of Sciences Nanjing College; Nanjing Institute of Software Technology), Ziyu Cui (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Qianwang Dai (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Jiansen Song (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Dong Wang (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Yu Gao (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Wei Wang (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences; University of Chinese Academy of Sciences Nanjing College; Nanjing Institute of Software Technology), Jun Wei (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences; University of Chinese Academy of Sciences Nanjing College; Nanjing Institute of Software Technology), Lei Chen (Inspur Software Group Co., Ltd., China), Hanmo Wang (Inspur Software Group Co., Ltd., China), Hua Zhong (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), and Tao Huang (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences)

SmallRace: Static Race Detection for Dynamic Languages - A Case on Smalltalk 1136
Siwei Cui (Texas A&M University, Texas), Yifei Gao (Texas A&M University, Texas), Rainer Unterguggenberger (Lam Research), Wilfried Pichler (Lam Research), Sean Livingstone (Texas A&M University, Texas), and Jeff Huang (Texas A&M University, Texas)

Studies on gender in SE

"STILL AROUND": Experiences and Survival Strategies of Veteran Women Software Developers 1148
Sterre van Breukelen (Eindhoven University of Technology, The Netherlands), Ann Barcomb (University of Calgary, Canada), Sebastian Baltes (University of Adelaide, Australia), and Alexander Serebrenik (Eindhoven University of Technology, The Netherlands)

AI testing 1

When and Why Test Generators for Deep Learning Produce Invalid Inputs: an Empirical Study .. 1161
Vincenzo Riccio (Università della Svizzera italiana, Switzerland) and Paolo Tonella (Università della Svizzera italiana, Switzerland)

Fuzzing Automatic Differentiation in Deep-Learning Libraries 1174
Chenyuan Yang (University of Illinois at Urbana-Champaign), Yinlin Deng (University of Illinois at Urbana-Champaign), Jiayi Yao (The Chinese University of Hong Kong, Shenzhen), Yuxing Tu (Huazhong University of Science and Technology), Hanchi Li (University of Science and Technology of China, China), and Lingming Zhang (University of Illinois at Urbana-Champaign)

Lightweight Approaches to DNN Regression Error Reduction: An Uncertainty Alignment Perspective 1187
Zenan Li (Nanjing University, China), Maorun Zhang (Nanjing University, China), Jingwei Xu (Nanjing University, China), Yuan Yao (Nanjing University, China), Chun Cao (Nanjing University, China), Taolue Chen (University of London, UK), Xiaoxing Ma (Nanjing University, China), and Jian Lü (Nanjing University, China)

Revisiting Neuron Coverage for DNN Testing: A Layer-Wise and Distribution-Aware Criterion .. 1200
Yuanyuan Yuan (The Hong Kong University of Science and Technology, China), Qi Pang (The Hong Kong University of Science and Technology, China), and Shuai Wang (The Hong Kong University of Science and Technology, China)

Code review

Code Review of Build System Specifications: Prevalence, Purposes, Patterns, and Perceptions 1213
Mahtab Nejati (University of Waterloo, Canada), Mahmoud Alfadel (University of Waterloo, Canada), and Shane McIntosh (University of Waterloo, Canada)

Program repair techniques and applications

Better Automatic Program Repair by Using Bug Reports and Tests Together 1225
Manish Motwani (University of Massachusetts, USA) and Yuriy Brun (University of Massachusetts, USA)

CCTEST: Testing and Repairing Code Completion Systems 1238
Zongjie Li (The Hong Kong University of Science and Technology, Hong Kong SAR), ChaoZheng Wang (Harbin Institute of Technology, China), Zhibo Liu (The Hong Kong University of Science and Technology, Hong Kong SAR), Haoxuan Wang (Swiss Federal Institute of Technology Lausanne, Switzerland), Dong Chen (The Hong Kong University of Science and Technology, Hong Kong SAR), Shuai Wang (The Hong Kong University of Science and Technology, Hong Kong SAR), and Cuiyun Gao (Harbin Institute of Technology, China)

KNOD: Domain Knowledge Distilled Tree Decoder for Automated Program Repair 1251
Nan Jiang (Purdue University, USA), Thibaud Lutellier (University of Alberta, Canada), Yiling Lou (Fudan University, China), Lin Tan (Purdue University, USA), Dan Goldwasser (Purdue University, USA), and Xiangyu Zhang (Purdue University, USA)

Rete: Learning Namespace Representation for Program Repair 1264
Nikhil Parasaram, Earl T. Barr, and Sergey Mechtaev

Requirements elicitation and understanding

AI-Based Question Answering Assistance for Analyzing Natural-Language Requirements 1277
Saad Ezzini (University of Luxembourg, Luxembourg), Sallam Abualhaija (University of Luxembourg, Luxembourg), Chetan Arora (Deakin University and Monash University, Australia), and Mehrdad Sabetzadeh (University of Ottawa, Canada)

Strategies, Benefits and Challenges of App Store-Inspired Requirements Elicitation 1290
Alessio Ferrari (Institute of Information Science and Technologies "A. Faedo", National Research Council (ISTI-CNR), Pisa, Italy) and Paola Spoletini (Kennesaw State University, USA)

Software verification

Data-Driven Recurrent Set Learning For Non-Termination Analysis 1303
Zhilei Han (Tsinghua University; Key Laboratory for Information System Security, MoE; Beijing National Research Center for Information Science and Technology, China) and Fei He (Tsinghua University; Key Laboratory for Information System Security, MoE; Beijing National Research Center for Information Science and Technology, China)

Compiling Parallel Symbolic Execution with Continuations 1316
Guannan Wei (Purdue University, USA), Songlin Jia (Purdue University, USA), Ruiqi Gao (Purdue University, USA), Haotian Deng (Purdue University, USA), Shangyin Tan (UC Berkeley, USA), Oliver Bracevac (Purdue University, USA), and Tiark Rompf (Purdue University, USA)

Verifying Data Constraint Equivalence in FinTech Systems 1329
Chengpeng Wang (The Hong Kong University of Science and Technology, China), Gang Fan (Ant Group, China), Peisen Yao (Zhejiang University, China), Fuxiong Pan (Ant Group, China), and Charles Zhang (The Hong Kong University of Science and Technology, China)

Tolerate Control-Flow Changes for Sound Data Race Prediction 1342
Shihao Zhu (State Key Laboratory of Computer Science, Institute of Software, Chinese Academy of Sciences and University of Chinese Academy of Sciences, China), Yuqi Guo (State Key Laboratory of Computer Science, Institute of Software, Chinese Academy of Sciences and University of Chinese Academy of Sciences, China), Long Zhang (State Key Laboratory of Computer Science, Institute of Software, Chinese Academy of Sciences and University of Chinese Academy of Sciences, China), and Yan Cai (State Key Laboratory of Computer Science, Institute of Software, Chinese Academy of Sciences and University of Chinese Academy of Sciences, China)

Testing of mobile, web and games

Fill in the Blank: Context-Aware Automated Text Input Generation for Mobile GUI Testing 1355
Zhe Liu (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Chunyang Chen (Monash University, Australia), Junjie Wang (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Xing Che (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Yuekai Huang (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Jun Hu (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), and Qing Wang (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China)

Detecting Dialog-Related Keyboard Navigation Failures in Web Applications 1368
Paul T. Chiou (University of Southern California, USA), Ali S. Alotaibi (University of Southern California, USA), and William G. J. Halfond (University of Southern California, USA)

COLUMBUS: Android App Testing Through Systematic Callback Exploration 1381
Priyanka Bose (University of California, Santa Barbara, USA), Dipanjan Das (University of California, Santa Barbara, USA), Saastha Vasan (University of California, Santa Barbara, USA), Sebastiano Mariani (VMware, Inc.), Ilya Grishchenko (University of California, Santa Barbara, USA), Andrea Continella (University of Twente), Antonio Bianchi (Purdue University), Christopher Kruegel (University of California, Santa Barbara, USA), and Giovanni Vigna (University of California, Santa Barbara, USA)

GameRTS: A Regression Testing Framework for Video Games 1393
Jiongchi Yu (Singapore Management University, Singapore), Yuechen Wu (NetEase Fuxi AI Lab, China), Xiaofei Xie (Singapore Management University, Singapore), Wei Le (Iowa State University, USA), Lei Ma (University of Alberta, Canada; The University of Tokyo, Japan), Yingfeng Chen (NetEase Fuxi AI Lab, China), Jingyu Hu (NetEase Fuxi AI Lab, China), and Fan Zhang (Zhejiang University, China; Zhengzhou Xinda Institute of Advanced Technology, China)

Recommender systems

Autonomy Is An Acquired Taste: Exploring Developer Preferences for GitHub Bots 1405
Amir Ghorbani (University of Victoria, Canada), Nathan Cassee (Eindhoven University of Technology, The Netherlands), Derek Robinson (University of Victoria, Canada), Adam Alami (Aalborg University, Denmark), Neil Ernst (University of Victoria, Canada), Alexander Serebrenik (Eindhoven University of Technology, The Netherlands), and Andrzej Wasowski (IT University of Copenhagen, Denmark)

Flexible and Optimal Dependency Management via Max-SMT 1418
Donald Pinckney (Northeastern University, USA), Federico Cassano (Northeastern University, USA), Arjun Guha (Northeastern University, USA), Jonathan Bell (Northeastern University, USA), Massimiliano Culpo (np-complete, S.r.l., Italy), and Todd Gamblin (Lawrence Livermore National Laboratory, USA)

Program repair with and for AI

Impact of Code Language Models on Automated Program Repair 1430
Nan Jiang (Purdue University, USA), Kevin Liu (Lynbrook High School, USA), Thibaud Lutellier (University of Alberta, USA), and Lin Tan (Purdue University, USA)

Tare: Type-Aware Neural Program Repair 1443
Qihao Zhu (Peking University, China), Zeyu Sun (Zhongguancun Laboratory, China), Wenjie Zhang (Peking University, China), Yingfei Xiong (Peking Univerisity, China), and Lu Zhang (Peking University, China)

Template-Based Neural Program Repair 1456
Xiangxin Meng (Beihang University, China), Xu Wang (Beihang University, China), Hongyu Zhang (Chongqing University, China), Hailong Sun (Beihang University, China), Xudong Liu (Beihang University, China), and Chunming Hu (Beihang University, China)

Automated Repair of Programs from Large Language Models 1469
Zhiyu Fan (National University of Singapore, Singapore), Xiang Gao (Beihang University, China), Martin Mirchev (National University of Singapore, Singapore), Abhik Roychoudhury (National University of Singapore, Singapore), and Shin Hwei Tan (Southern University of Science and Technology, China)

Automated Program Repair in the Era of Large Pre-Trained Language Models 1482
Chunqiu Steven Xia (University of Illinois Urbana-Champaign), Yuxiang Wei (University of Illinois Urbana-Champaign), and Lingming Zhang (University of Illinois Urbana-Champaign)

Programming languages

Faster or Slower? Performance Mystery of Python Idioms Unveiled with Empirical Evidence 1495
Zejun Zhang (Australian National University, Australia), Zhenchang Xing (Data61, CSIRO, Australia; Australian National University, Australia), Xin Xia (Software Engineering Application Technology Lab, China), Xiwei Xu (Data61, CSIRO, Australia), Liming Zhu (Data61, CSIRO, Australia), and Qinghua Lu (Data61, CSIRO, Australia)

Testability Refactoring in Pull Requests: Patterns and Trends 1508
Pavel Reich (University of Hamburg, Germany) and Walid Maalej (University of Hamburg, Germany)

Usability-Oriented Design of Liquid Types for Java 1520
Catarina Gamboa (Carnegie Mellon University, USA, and LASIGE, Faculdade de Ciências da Universidade de Lisboa, Portugal), Paulo Canelas (Carnegie Mellon University, USA, and LASIGE, Faculdade de Ciências da Universidade de Lisboa, Portugal), Christopher S. Timperley (Carnegie Mellon University, USA), and Alcides Fonseca (LASIGE, Faculdade de Ciências da Universidade de Lisboa, Portugal)

AI bias and fairness

Towards Understanding Fairness and its Composition in Ensemble Machine Learning 1533
Usman Gohar (Iowa State University, USA), Sumon Biswas (Carnegie Mellon University, USA), and Hridesh Rajan (Iowa State University, USA)

Fairify: Fairness Verification of Neural Networks 1546
Sumon Biswas (Carnegie Mellon University, USA) and Hridesh Rajan (Iowa State University, USA)

Leveraging Feature Bias for Scalable Misprediction Explanation of Machine Learning Models 1559
Jiri Gesi (University of California, USA), Xinyun Shen (University of California, USA), Yunfan Geng (University of California, USA), Qihong Chen (University of California, USA), and Iftekhar Ahmed (University of California, USA)

Information-Theoretic Testing and Debugging of Fairness Defects in Deep Neural Networks 1571
Verya Monjezi (University of Texas at El Paso), Ashutosh Trivedi (University of Colorado Boulder), Gang Tan (Pennsylvania State University), and Saeid Tizpaz-Niari (University of Texas at El Paso)

Requirements engineering

Demystifying Privacy Policy of Third-Party Libraries in Mobile Apps 1583
Kaifa Zhao (The Hong Kong Polytechnic University), Xian Zhan (Southern University of Science and Technology), Le Yu (The Hong Kong Polytechnic University), Shiyao Zhou (The Hong Kong Polytechnic University), Hao Zhou (The Hong Kong Polytechnic University), Xiapu Luo (The Hong Kong Polytechnic University), Haoyu Wang (Huazhong University of Science and Technology), and Yepang Liu (Southern University of Science and Technology)

Cross-Domain Requirements Linking via Adversarial-Based Domain Adaptation 1596
Zhiyuan Chang (Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Mingyang Li (Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Qing Wang (Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Shoubin Li (Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), and Junjie Wang (Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China)

SE for security 2

On-Demand Security Requirements Synthesis with Relational Generative Adversarial Networks . 1609
Viktoria Koscinski (Rochester Institute of Technology, USA), Sara Hashemi (Rochester Institute of Technology, USA), and Mehdi Mirakhorli (Rochester Institute of Technology, USA)

Measuring Secure Coding Practice and Culture: A Finger Pointing at the Moon is not the Moon 1622
Ita Ryan (University College Cork, Ireland), Utz Roedig (University College Cork, Ireland), and Klaas-Jan Stol (University College Cork, Ireland)

What Challenges Do Developers Face About Checked-in Secrets in Software Artifacts? 1635
Setu Kumar Basak (North Carolina State University, USA), Lorenzo Neil (North Carolina State University, USA), Bradley Reaves (North Carolina State University, USA), and Laurie Williams (North Carolina State University, USA)

Lejacon: A Lightweight and Efficient Approach to Java Confidential Computing on SGX 1648
Xinyuan Miao (Shanghai Jiao Tong University, China), Ziyi Lin (Alibaba Group, China), Shaojun Wang (Alibaba Group, China), Lei Yu (Alibaba Group, China), Sanhong Li (Alibaba Group, China), Zihan Wang (Shanghai Jiao Tong University, China), Pengbo Nie (Shanghai Jiao Tong University, China), Yuting Chen (Shanghai Jiao Tong University, China), Beijun Shen (Shanghai Jiao Tong University, China), and He Jiang (Dalian University of Technology, China)

Keyword Extraction From Specification Documents for Planning Security Mechanisms 1661
Jeffy Jahfar Poozhithara (University of Washington Bothell, USA), Hazeline U. Asuncion (University of Washington Bothell, USA), and Brent Lagesse (University of Washington Bothell, USA)

Software Evolution

Dependency Facade: The Coupling and Conflicts Between Android Framework and Its Customization 1674
Wuxia Jin (Xi'an Jiaotong University, China), Yitong Dai (Xi'an Jiaotong University, China), Jianguo Zheng (Xi'an Jiaotong University, China), Yu Qu (University of California, United States), Ming Fan (Xi'an Jiaotong University, China), Zhenyu Huang (Honor Device Co., Ltd., China), Dezhi Huang (Honor Device Co., Ltd., China), and Ting Liu (Xi'an Jiaotong University, China)

Test quality and improvement

Test Selection for Unified Regression Testing 1687
Shuai Wang (University of Illinois at Urbana-Champaign, USA), Xinyu Lian (University of Illinois at Urbana-Champaign, USA), Darko Marinov (University of Illinois at Urbana-Champaign, USA), and Tianyin Xu (University of Illinois at Urbana-Champaign, USA)

ATM: Black-box Test Case Minimization Based on Test Code Similarity and Evolutionary Search 1700
Rongqi Pan (University of Ottawa, Canada), Taher A. Ghaleb (University of Ottawa, Canada), and Lionel Briand (University of Ottawa, Canada; University of Luxembourg, Luxembourg)

Measuring and Mitigating Gaps in Structural Testing 1712
Soneya Binta Hossain (University of Virginia), Matthew B. Dwyer (University of Virginia), Sebastian Elbaum (University of Virginia), and Anh Nguyen-Tuong (University of Virginia)

Runtime analysis and self-adaptation

Heterogeneous Anomaly Detection for Software Systems via Semi-Supervised Cross-Modal Attention 1724
Cheryl Lee (The Chinese University of Hong Kong, China), Tianyi Yang (The Chinese University of Hong Kong, China), Zhuangbin Chen (The Chinese University of Hong Kong, China), Yuxin Su (Sun Yat-sen University, China), Yongqiang Yang (Huawei Cloud, China), and Michael R. Lyu (The Chinese University of Hong Kong, China)

Recommending Root-Cause and Mitigation Steps for Cloud Incidents Using Large Language Models 1737
Toufique Ahmed (UC Davis), Supriyo Ghosh (Microsoft), Chetan Bansal (Microsoft), Thomas Zimmermann (Microsoft Research), Xuchao Zhang (Microsoft), and Saravan Rajmohan (Microsoft)

Eadro: An End-to-End Troubleshooting Framework for Microservices on Multi-Source Data 1750
Cheryl Lee (The Chinese University of Hong Kong, China), Tianyi Yang (The Chinese University of Hong Kong, China), Zhuangbin Chen (The Chinese University of Hong Kong, China), Yuxin Su (Sun Yat-sen University, China), and Michael R. Lyu (The Chinese University of Hong Kong, China)

LogReducer: Identify and Reduce Log Hotspots in Kernel on the Fly 1763
Guangba Yu (Sun Yat-Sen University, China), Pengfei Chen (Sun Yat-Sen University, China), Pairui Li (Tencent Inc., China), Tianjun Weng (Tencent Inc., China), Haibing Zheng (Tencent Inc., China), Yuetang Deng (Tencent Inc., China), and Zibin Zheng (Sun Yat-sen University, China)

AI testing 2

Aries: Efficient Testing of Deep Neural Networks via Labeling-Free Accuracy Estimation 1776
Qiang Hu (University of Luxembourg, Luxembourg), Yuejun Guo (Luxembourg Institute of Science and Technology, Luxembourg), Xiaofei Xie (Singapore Management University, Singapore), Maxime Cordy (University of Luxembourg, Luxembourg), Lei Ma (University of Alberta, Canada; The University of Tokyo, Japan), Mike Papadakis (University of Luxembourg, Luxembourg), and Yves Le Traon (University of Luxembourg, Luxembourg)

CC: Causality-Aware Coverage Criterion for Deep Neural Networks 1788
Zhenlan Ji (The Hong Kong University of Science and Technology, China), Pingchuan Ma (The Hong Kong University of Science and Technology, China), Yuanyuan Yuan (The Hong Kong University of Science and Technology, China), and Shuai Wang (The Hong Kong University of Science and Technology, China)

Balancing Effectiveness and Flakiness of Non-Deterministic Machine Learning Tests 1801
Chunqiu Steven Xia (University of Illinois Urbana-Champaign), Saikat Dutta (University of Illinois Urbana-Champaign), Sasa Misailovic (University of Illinois Urbana-Champaign), Darko Marinov (University of Illinois Urbana-Champaign), and Lingming Zhang (University of Illinois Urbana-Champaign)

Many-Objective Reinforcement Learning for Online Testing of DNN-Enabled Systems 1814
Fitash Ul Haq (University of Luxembourg, Luxembourg), Donghwan Shin (University of Sheffield, United Kingdom), and Lionel C. Briand (University of Luxembourg, Luxembourg; University of Ottawa, Canada)

Reliability Assurance for Deep Neural Network Architectures Against Numerical Defects 1827
Linyi Li (University of Illinois Urbana-Champaign), Yuhao Zhang (University of Wisconsin-Madison), Luyao Ren (Peking University), Yingfei Xiong (Peking University), and Tao Xie (Peking University)

Developers' forums

Demystifying Issues, Challenges, and Solutions for Multilingual Software Development 1840
Haoran Yang (Washington State University, USA), Weile Lian (Washington State University, USA), Shaowei Wang (University of Manitoba, Canada), and Haipeng Cai (Washington State University, USA)

Automated Summarization of Stack Overflow Posts 1853
Bonan Kou (Purdue University, USA), Muhao Chen (University of Sourthern California, USA), and Tianyi Zhang (Purdue University, USA)

Semi-Automatic, Inline and Collaborative Web Page Code Curations 1866
Roy Rutishauser (University of Zurich, Switzerland), André N. Meyer (University of Zurich, Switzerland), Reid Holmes (University of British Columbia, Canada), and Thomas Fritz (University of Zurich, Switzerland)

Program comprehension

Identifying Key Classes for Initial Software Comprehension: Can We Do It Better? 1878
Weifeng Pan (Zhejiang Gongshang University, China), Xin Du (Zhejiang Gongshang University, China), Hua Ming (Oakland University, USA), Dae-Kyoo Kim (Oakland University, USA), and Zijiang Yang (Xi'an Jiaotong University, China)

Improving API Knowledge Discovery with ML: A Case Study of Comparable API Methods 1890
Daye Nam (Carnegie Mellon University, USA), Brad Myers (Carnegie Mellon University, USA), Bogdan Vasilescu (Carnegie Mellon University, USA), and Vincent Hellendoorn (Carnegie Mellon University, USA)

Evidence Profiles for Validity Threats in Program Comprehension Experiments 1907
Marvin Muñoz Barón (University of Stuttgart, Germany), Marvin Wyrich (University of Stuttgart, Germany), Daniel Graziotin (University of Stuttgart, Germany), and Stefan Wagner (University of Stuttgart, Germany)

Developers' Visuo-Spatial Mental Model and Program Comprehension 1920
Abir Bouraffa (University of Hamburg, Germany), Gian-Luca Fuhrmann (University of Hamburg, Germany), and Walid Maalej (University of Hamburg, Germany)

Two Sides of the Same Coin: Exploiting the Impact of Identifiers in Neural Code Comprehension 1933
Shuzheng Gao (Harbin Institute of Technology, China), Cuiyun Gao (Harbin Institute of Technology, China), Chaozheng Wang (Harbin Institute of Technology, China), Jun Sun (Singapore Management University, Singapore), David Lo (Singapore Management University, Singapore), and Yue Yu (National University of Defense Technology)

Reverse engineering

SeeHow: Workflow Extraction from Programming Screencasts Through Action-Aware Video Analytics 1946
Dehai Zhao (Australian National University and CSIRO's data61, Australia), Zhenchang Xing (Australian National University and CSIRO's data61, Australia), Xin Xia (Software Engineering Application Technology Lab, Huawei, China), Deheng Ye (Tencent AI Lab, China), Xiwei Xu (CSIRO's data61, Australia), and Liming Zhu (CSIRO's data61, Australia)

AidUI: Toward Automated Recognition of Dark Patterns in User Interfaces 1958
SM Hasan Mansur (George Mason University), Sabiha Salma (George Mason University), Damilola Awofisayo (Duke University), and Kevin Moran (George Mason University)

Carving UI Tests to Generate API Tests and API Specification 1971
Rahul Yandrapally (University of British Columbia, Canada), Saurabh Sinha (IBM Research, USA), Rachel Tzoref-Brill (IBM Research, Israel), and Ali Mesbah (University of British Columbia, Canada)

Ex Pede Herculem: Augmenting Activity Transition Graph for Apps via Graph Convolution Network 1983
Zhe Liu (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Chunyang Chen (Monash University), Junjie Wang (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Yuhui Su (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Yuekai Huang (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), Jun Hu (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China), and Qing Wang (State Key Laboratory of Intelligent Game, China; Science and Technology on Integrated Information System Laboratory, Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences, China)

Software processes

Sustainability is Stratified: Toward a Better Theory of Sustainable Software Engineering 1996
Sean McGuire (Dalhousie University, Canada), Erin Shultz (Dalhousie University, Canada), Bimpe Ayoola (Dalhousie University, Canada), and Paul Ralph (Dalhousie University, Canada)

Static analysis

DLInfer: Deep Learning with Static Slicing for Python Type Inference 2009
Yanyan Yan (Nanjing University, China), Yang Feng (Nanjing University, China), Hongcheng Fan (Nanjing University, China), and Baowen Xu (Nanjing University, China)

ViolationTracker: Building Precise Histories for Static Analysis Violations 2022
Ping Yu (Fudan University, China; Shanghai Key Laboratory of Data Science, China), Yijian Wu (Fudan University, China; Shanghai Key Laboratory of Data Science, China), Xin Peng (Fudan University, China; Shanghai Key Laboratory of Data Science, China), Jiahan Peng (Fudan University, China; Shanghai Key Laboratory of Data Science, China), Jian Zhang (Fudan University, China; Shanghai Key Laboratory of Data Science, China), Peicheng Xie (Fudan University, China; Shanghai Key Laboratory of Data Science, China), and Wenyun Zhao (Fudan University, China; Shanghai Key Laboratory of Data Science, China)

Testing of database and low-level software

Compiler Test-Program Generation via Memoized Configuration Search 2035
Junjie Chen (Tianjin University, China), Chenyao Suo (Tianjin University, China), Jiajun Jiang (Tianjin University, China), Peiqi Chen (Tianjin University, China), and Xingjian Li (Tianjin University, China)

Generating Test Databases for Database-Backed Applications 2048
Cong Yan (Microsoft Research, USA), Suman Nath (Microsoft Research, USA), and Shan Lu (University of Chicago, USA)

Testing Database Engines via Query Plan Guidance 2060
Jinsheng Ba (National University of Singapore, Singapore) and Manuel Rigger (National University of Singapore, Singapore)

Testing Database Systems via Differential Query Execution 2072
Song Jiansen (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Dou Wensheng (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences; University of Chinese Academy of Sciences Nanjing College; Nanjing Institute of Software Technology), Cui Ziyu (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Dai Qianwang (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Wang Wei (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences; University of Chinese Academy of Sciences Nanjing College; Nanjing Institute of Software Technology), Wei Jun (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences; University of Chinese Academy of Sciences Nanjing College; Nanjing Institute of Software Technology), Zhong Hua (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), and Huang Tao (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences)

Software performance

Analyzing the Impact of Workloads on Modeling the Performance of Configurable Software Systems 2085
Stefan Mühlbauer (Leipzig University, Germany), Florian Sattler (Saarland University, Saarland Informatics Campus, Germany), Christian Kaltenecker (Saarland University, Saarland Informatics Campus, Germany), Johannes Dorn (Leipzig University, Germany), Sven Apel (Saarland University, Saarland Informatics Campus, Germany), and Norbert Siegmund (Leipzig University, ScaDS.AI Dresden/Leipzig, Germany)

Twins or False Friends? A Study on Energy Consumption and Performance of Configurable Software 2098
Max Weber (Leipzig University, Germany), Christian Kaltenecker (Saarland University, Saarland Informatics Campus, Germany), Florian Sattler (Saarland University, Saarland Informatics Campus, Germany), Sven Apel (Saarland University, Saarland Informatics Campus, Germany), and Norbert Siegmund (Leipzig University, Germany)

Code generation

Learning Deep Semantics for Test Completion 2111
Pengyu Nie (The University of Texas at Austin, USA), Rahul Banerjee (The University of Texas at Austin, USA), Junyi Jessy Li (The University of Texas at Austin, USA), Raymond J. Mooney (The University of Texas at Austin, USA), and Milos Gligoric (The University of Texas at Austin, USA)

SkCoder: A Sketch-Based Approach for Automatic Code Generation 2124
Jia Li (Peking University, China), Yongmin Li (Peking University, China), Ge Li (Peking University, China), Zhi Jin (Peking University, China), Yiyang Hao (aiXcoder, China), and Xing Hu (Zhejiang University, China)

An Empirical Comparison of Pre-Trained Models of Source Code 2136
Changan Niu (Nanjing University, China), Chuanyi Li (Nanjing University, China), Vincent Ng (University of Texas at Dallas, USA), Dongxiao Chen (Nanjing University, China), Jidong Ge (Nanjing University), and Bin Luo (Nanjing University, China)

On the Robustness of Code Generation Techniques: An Empirical Study on GitHub Copilot 2149
Antonio Mastropaolo (Università della Svizzera italiana, Switzerland), Luca Pascarella (Università della Svizzera italiana, Switzerland), Emanuela Guglielmi (University of Molise, Italy), Matteo Ciniselli (Università della Svizzera italiana, Switzerland), Simone Scalabrino (University of Molise, Italy), Rocco Oliveto (University of Molise, Italy), and Gabriele Bavota (Università della Svizzera italiana, Switzerland)

Source Code Recommender Systems: The Practitioners' Perspective 2161
Matteo Ciniselli (Università della Svizzera italiana, Switzerland), Luca Pascarella (Università della Svizzera italiana, Switzerland), Emad Aghajani (Università della Svizzera italiana, Switzerland), Simone Scalabrino (University of Molise, Italy), Rocco Oliveto (University of Molise, Italy), and Gabriele Bavota (Università della Svizzera italiana, Switzerland)

Software development tools

Safe Low-Level Code Without Overhead is Practical 2173
Solal Pirelli (EPFL) and George Candea (EPFL)

Sibyl: Improving Software Engineering Tools with SMT Selection 2185
Will Leeson (University of Virginia, USA), Matthew B. Dwyer (University of Virginia, USA), and Antonio Filieri (Imperial College London, UK)

CoCoSoDa: Effective Contrastive Learning for Code Search 2198
Ensheng Shi (Xi'an Jiaotong University, China), Yanlin Wang (Sun Yat-sen University, China), Wenchao Gu (The Chinese University of Hong Kong, China), Lun Du (Microsoft Research, China), Hongyu Zhang (The University of Newcastle, Austrialia), Shi Han (Microsoft Research, China), Dongmei Zhang (Microsoft Research, China), and Hongbin Sun (Xi'an Jiaotong University, China)

Fault injection and mutation

Coverage Guided Fault Injection for Cloud Systems 2211
Yu Gao (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Wensheng Dou (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences; University of Chinese Academy of Sciences Nanjing College; Nanjing Institute of Software Technology), Dong Wang (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Wenhan Feng (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), Jun Wei (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences; University of Chinese Academy of Sciences Nanjing College; Nanjing Institute of Software Technology), Hua Zhong (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences), and Tao Huang (Institute of Software Chinese Academy of Sciences, China; University of Chinese Academy of Sciences)

Diver: Oracle-Guided SMT Solver Testing with Unrestricted Random Mutations 2224
Jongwook Kim (Korea University, Republic of Korea), Sunbeom So (Korea University, Republic of Korea), and Hakjoo Oh (Korea University, Republic of Korea)

Vulnerability detection

An Empirical Study of Deep Learning Models for Vulnerability Detection 2237
Benjamin Steenhoek (Iowa State University, USA), Md Mahbubur Rahman (Iowa State University, USA), Richard Jiles (Iowa State University, USA), and Wei Le (Iowa State University, USA)

DeepVD: Toward Class-Separation Features for Neural Network Vulnerability Detection 2249
Wenbo Wang (New Jersey Institute of Technology, USA), Tien N. Nguyen (University of Texas at Dallas, USA), Shaohua Wang (New Jersey Institute of Technology, USA), Yi Li (New Jersey Institute of Technology, USA), Jiyuan Zhang (University of Illinois Urbana-Champaign, USA), and Aashish Yadavally (University of Texas at Dallas, USA)

Enhancing Deep Learning-Based Vulnerability Detection by Building Behavior Graph Model 2262
Bin Yuan (Huazhong University of Science and Technology, China; Shenzhen Huazhong University of Science and Technology Research Institute, China), Yifan Lu (Huazhong University of Science and Technology, China), Yilin Fang (Huazhong University of Science and Technology, China), Yueming Wu (Huazhong University of Science and Technology, China), Deqing Zou (Huazhong University of Science and Technology, China), Zhen Li (Huazhong University of Science and Technology, China), Zhi Li (Huazhong University of Science and Technology, China), and Hai Jin (Huazhong University of Science and Technology, China)

Vulnerability Detection with Graph Simplification and Enhanced Graph Representation Learning 2275
Xin-Cheng Wen (Harbin Institute of Technology, China), Yupan Chen (Harbin Institute of Technology, China), Cuiyun Gao (Harbin Institute of Technology, China), Hongyu Zhang (Chongqing University, China), Jie M. Zhang (King's College London, UK), and Qing Liao (Harbin Institute of Technology, China)

Does Data Sampling Improve Deep Learning-Based Vulnerability Detection? Yeas! and Nays! 2287
Xu Yang (University of Manitoba, Canada), Shaowei Wang (University of Manitoba, Canada), Yi Li (New Jersey Institute of Technology, USA), and Shaohua Wang (New Jersey Institute of Technology, USA)

Issue reporting and reproduction

Incident-Aware Duplicate Ticket Aggregation for Cloud Systems 2299
Jinyang Liu (The Chinese University of Hong Kong, China), Shilin He (Microsoft Research, China), Zhuangbin Chen (The Chinese University of Hong Kong, China), Liqun Li (Microsoft Research, China), Yu Kang (Microsoft Research, China), Xu Zhang (Microsoft Research, China), Pinjia He (The Chinese University of Hong Kong, China), Hongyu Zhang (Chongqing University, China), Qingwei Lin (Microsoft Research, China), Zhangwei Xu (Microsoft Azure, USA), Saravan Rajmohan (Microsoft 365, USA), Dongmei Zhang (Microsoft Research, China), and Michael R. Lyu (The Chinese University of Hong Kong, China)

Large Language Models are Few-Shot Testers: Exploring LLM-Based General Bug Reproduction . 2312
Sungmin Kang (Korea Advanced Institute of Science and Technology, Republic of Korea), Juyeon Yoon (Korea Advanced Institute of Science and Technology, Republic of Korea), and Shin Yoo (Korea Advanced Institute of Science and Technology, Republic of Korea)

On the Reproducibility of Software Defect Datasets 2324
Hao-Nan Zhu (University of California, United States of America) and Cindy Rubio-González (University of California, United States of America)

Context-Aware Bug Reproduction for Mobile Apps 2336
Yuchao Huang (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Junjie Wang (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Zhe Liu (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), Song Wang (York University, Canada), Chunyang Chen (Monash University, Australia), Mingyang Li (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China), and Qing Wang (Institute of Software Chinese Academy of Sciences, China; State Key Laboratory of Intelligent Game, China; University of Chinese Academy of Sciences, China)

Read It, Don't Watch It: Captioning Bug Recordings Automatically 2349
Sidong Feng (Monash University, Australia), Mulong Xie (Australian National University, Australia), Yinxing Xue (University of Science and Technology of China, China), and Chunyang Chen (Monash University, Australia)

Software quality

DuetCS: Code Style Transfer Through Generation and Retrieval 2362
Binger Chen (Technische Universität Berlin, Germany) and Ziawasch Abedjan (Leibniz Universität Hannover, Germany)

SE education methods and tools

On the Applicability of Language Models to Block-Based Programs 2374
Elisabeth Griebl (University of Passau, Germany), Benedikt Fein (University of Passau, Germany), Florian Obermüller (University of Passau, Germany), Gordon Fraser (University of Passau, Germany), and René Just (University of Washington, USA)

Metamorphic testing

MTTM: Metamorphic Testing for Textual Content Moderation Software 2387
Wenxuan Wang (The Chinese University of Hong Kong, China), Jen-tse Huang (The Chinese University of Hong Kong, China), Weibin Wu (Sun Yat-sen University, China), Jianping Zhang (The Chinese University of Hong Kong, China), Yizhan Huang (The Chinese University of Hong Kong, China), Shuqing Li (The Chinese University of Hong Kong, China), Pinjia He (The Chinese University of Hong Kong, Shenzhen, China), and Michael R. Lyu (The Chinese University of Hong Kong, China)

Metamorphic Shader Fusion for Testing Graphics Shader Compilers 2400
Dongwei Xiao (The Hong Kong University of Science and Technology, China), Zhibo Liu (Hong Kong University of Science and Technology, China), and Shuai Wang (Hong Kong University of Science and Technology, China)

MorphQ: Metamorphic Testing of the Qiskit Quantum Computing Platform 2413
Matteo Paltenghi (University of Stuttgart, Germany) and Michael Pradel (University of Stuttgart, Germany)

Pre-trained and few shot learning for SE

Automating Code-Related Tasks Through Transformers: The Impact of Pre-Training 2425
Rosalia Tufano (Università della Svizzera italiana, Switzerland), Luca Pascarella (Università della Svizzera italiana, Switzerland), and Gabriele Bavota (Università della Svizzera italiana, Switzerland)

Log Parsing with Prompt-Based Few-Shot Learning 2438
Van-Hoang Le (The University of Newcastle, Australia) and Hongyu Zhang (Chongqing University, China)

Retrieval-Based Prompt Selection for Code-Related Few-Shot Learning 2450
Noor Nashid (University of British Columbia, Canada), Mifta Sintaha (University of British Columbia, Canada), and Ali Mesbah (University of British Columbia, Canada)

An Empirical Study of Pre-Trained Model Reuse in the Hugging Face Deep Learning Model Registry 2463
Wenxin Jiang (Purdue University, USA), Nicholas Synovic (Loyola University Chicago, USA), Matt Hyatt (Loyola University Chicago, USA), Taylor R. Schorlemmer (Purdue University, USA), Rohan Sethi (Loyola University Chicago, USA), Yung-Hsiang Lu (Purdue University, USA), George K. Thiruvathukal (Loyola University Chicago, USA), and James C. Davis (Purdue University, USA)

ContraBERT: Enhancing Code Pre-trained Models via Contrastive Learning 2476
Liu Shangqing (Nanyang Technological University, Singapore), Wu Bozhi (Nanyang Technological University, Singapore), Xie Xiaofei (Singapore Management University, Singapore), Meng Guozhu (SKLOIS, Institute of Information Engineering, Chinese Academy of Sciences, China), and Liu Yang (Nanyang Technological University, Singapore)

Program analysis

DStream: A Streaming-Based Highly Parallel IFDS Framework 2488
Xizao Wang (Nanjing University, China), Zhiqiang Zuo (Nanjing University, China), Lei Bu (Nanjing University, China), and Jianhua Zhao (Nanjing University, China)

(Partial) Program Dependence Learning 2501
Aashish Yadavally (University of Texas at Dallas, USA), Tien N. Nguyen (University of Texas at Dallas, USA), Wenbo Wang (New Jersey Institute of Technology, USA), and Shaohua Wang (New Jersey Institute of Technology, USA)

MirrorTaint: Practical Non-Intrusive Dynamic Taint Tracking for JVM-Based Microservice Systems 2514
Yicheng Ouyang (University of Illinois Urbana-Champaign, USA), Kailai Shao (Ant Group, China), Kunqiu Chen (Southern University of Science and Technology, China), Ruobing Shen (Peking University, China), Chao Chen (Ant Group, China), Mingze Xu (Ant Group, China), Yuqun Zhang (Southern University of Science and Technology, China), and Lingming Zhang (University of Illinois Urbana-Champaign, USA)

Vulnerability testing and patching

VulGen: Realistic Vulnerability Generation via Pattern Mining and Deep Learning 2527
Yu Nong (Washington State University, USA), Yuzhe Ou (The University of Texas at Dallas, USA), Michael Pradel (University of Stuttgart, Germany), Feng Chen (The University of Texas at Dallas, USA), and Haipeng Cai (Washington State University, USA)

Compatible Remediation on Vulnerabilities from Third-Party Libraries for Java Projects 2540
Lyuye Zhang (Nanyang Technological University, Singapore), Chengwei Liu (Nanyang Technological University, Singapore), Zhengzi Xu (Nanyang Technological University, Singapore), Sen Chen (Tianjin University, China), Lingling Fan (Nankai University, China), Lida Zhao (Nanyang Technological University, Singapore), Jiahui Wu (Nanyang Technological University, Singapore), and Yang Liu (Nanyang Technological University, Singapore)

Automated Black-box Testing of Mass Assignment Vulnerabilities in RESTful APIs 2553
Davide Corradini (Univerity of Verona, Italy), Michele Pasqua (University of Verona, Italy), and Mariano Ceccato (University of Verona, Italy)

CoLeFunDa: Explainable Silent Vulnerability Fix Identification 2565
Jiayuan Zhou (Huawei, Canada), Michael Pacheco (Huawei, Canada), Jinfu Chen (Huawei, Canada), Xing Hu (Zhejiang University, China), Xin Xia (Huawei, China), David Lo (Singapore Management University, Singapore), and Ahmed E. Hassan (Queen's University, Canada)

Cyber-physical systems testing

Finding Causally Different Tests for an Industrial Control System 2578
Christopher M. Poskitt (Singapore Management University, Singapore), Yuqi Chen (ShanghaiTech University, China; Shanghai Engineering Research Center of Energy Efficient and Custom AI IC, China), Jun Sun (Singapore Management University, Singapore), and Yu Jiang (Tsinghua University, China)

Doppelganger Test Generation for Revealing Bugs in Autonomous Driving Software 2591
Yuqi Huai (University of California, Irvine), Yuntianyi Chen (University of California, Irvine), Sumaya Almanee (University of California, Irvine), Tuan Ngo (VNU University of Engineering and Technology), Xiang Liao (University of California, Irvine), Ziwen Wan (University of California, Irvine), Qi Alfred Chen (University of California, Irvine), and Joshua Garcia (University of California, Irvine)

Generating Realistic and Diverse Tests for LiDAR-Based Perception Systems 2604
Garrett Christian (University of Virginia, USA), Trey Woodlief (University of Virginia, USA), and Sebastian Elbaum (University of Virginia, USA)

Software ecosystems

Rules of Engagement: Why and How Companies Participate in OSS 2617
Mariam Guizani (Oregon State University, USA), Aileen Abril Castro-Guzman (Oregon State University, USA), Anita Sarma (Oregon State University, USA), and Igor Steinmacher (Northern Arizona University, USA)

An Empirical Study on Software Bill of Materials: Where We Stand and the Road Ahead 2630
Boming Xia (CSIRO's Data61, Australia; University of New South Wales, Australia), Tingting Bi (CSIRO's Data61, Australia; Monash University, Australia), Zhenchang Xing (CSIRO's Data61, Australia; Australian National University, Australia), Qinghua Lu (CSIRO's Data61, Australia; University of New South Wales, Australia), and Liming Zhu (CSIRO's Data61, Australia; University of New South Wales, Australia)

Author Index **2643**

Message from the ICSE 2023 General Chair

Welcome to the 45th edition of the IEEE/ACM International Conference on Software Engineering - ICSE2023! It has been my great privilege to act as General Chair for this conference - again in somewhat challenging times, even if supposedly "post-pandemic". ICSE 2023 is a primarily in-person meeting, though we needed to provide remote presentation support for a number of authors unable to attend the conference in person. The Melbourne Convention and Exhibition Centre is designed as an in-person venue and while live present/Q&A and livestream can be supported, this requires considerable extra support and scheduling of the programme. So saying, ICSE 2020 to ICSE 2022 showed the potential of virtual attendance e.g. for those unable to travel. This was the second time ICSE has been held in Melbourne, Australia, the previous being in 1992.

ICSE is the leading and by far the largest conference in Software Engineering, attracting researchers, practitioners and students from around the world. ICSE2023 is co-located with 10 conferences and symposia this year, many long-established and prestigious venues in their own right. These include:

- The 4th ACM/IEEE International Conference on Automation of Software Test (AST 2023)
- The 2nd Conference on AI Engineering Software Engineering for AI (CAIN 2023)
- 16th International Conference on Cooperative and Human Aspects of Software Engineering (CHASE 2023)
- FormaliSE 2023
- 31st IEEE/ACM International Conference on Program Comprehension (ICPC 2023)
- 17th International Conference on Software and System Processes (ICSSP 2023) - welcoming back to co-location with ICSE!!
- 10th International Conference on Mobile Software Engineering and Systems 2023 (MobileSoft 2023)
- 20th Mining Software Repositories Conference (MSR 2023)
- International Conference on Technical Debt 2023
- 18th Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS 2023)

In addition, 22 workshops are co-located with ICSE2023, ⅔ running the three days before the conference and ⅓ on the Saturday after the main conference concluded. A few workshops and co-located conferences are running virtual sessions in addition to hybrid sessions. ICSE again hosts a Doctoral Symposium with over 20 PhD students presenting; a student research competition; New Faculty Symposium, Artefacts track, SCORE student competition, Student Research competition, and student mentoring programme. We were also fortunate to have seven technical briefings accepted and presented during the main conference three days.

Special programme items this year include an ICSE Showcase, where invited presentations from several specialised software engineering conferences have been sought to address some of the - historically important and popular - areas of SE that have in recent times had very few papers appearing at ICSE.This includes Requirements Engineering, modelling, software architecture and formal methods focused papers. We hope via the showcase track to spur increased submissions and interest from these communities in ICSE. A Future of Software Engineering invited speakers track highlights a number of directions in SE over the next 5-10+ years from leading established and emerging researchers in these areas. A post-ICSE Future of Software Engineering proceedings is planned to capture not only these speakers' visions, but also feedback from attendees from their panel-oriented presentation and discussion sessions. An Industry Forum aimed at local industry participation attracted a good number of submissions by industry practitioners, with optional short paper in proceedings.

A wide variety of diversity and social events have been organised. These include but are not limited to:

- Newcomers to ICSE
- Women@ICSE Lunch
- LGBTQI+@ICSE
- BIPOC@ICSE
- Conversation Café on Equality, Diversity and Inclusion (EDI)
- Conversation Café on Wellness in Academia
- Book Club discussion
- ICSE Run
- ICSE Tennis
- Recruitment database
- Recruitment lunch

Sustainability of conferences - and software engineering's contribution to sustainability - has become an important consideration. One issue with running ICSE in Australia is the distance to travel and length of stays needed for international participants. We highlighted carbon footprint relating to travel, chose as our venue the Melbourne Conference and Exhibition Centre due to its world-leading conference facility energy, food and conference support considerations, selected local hotels that have demonstrated high commitment to all aspects of sustainability, highlighted public transport options close to and far from the venue, and enabled remote presentation without need to attend ICSE in person for those needing - or choosing - to do so.

I would like to thank my Programme Co-chairs, Lori Pollock and Massimiliano Di Penta, for the huge efforts they have dedicated to not just running the Technical Programme track but

organising main conference track sessions, many contributions to overall ICSE 2023 decision making, and being great fun to work with. Christoph Treude has been a marvellous Treasurer under very challenging circumstances. Omar Haggag has gone well above and beyond the call of duty on "ICSE Web Chair" responsibilities and has provided leadership and support in many areas of ICSE organisation. Thanks to Sam Malek and Dongmei Zhang as workshop co-chairs, and Anna Liu and Henry Muccini for running the SEIP track. I want to thank Gabriele Bavota and Dan Hao for running the NIER track, Sonia Haiduc and Jean-Guy Schneider who chaired the SEET track, Rashina Hoda and Alexander Serebrenik the SEIS Track, and Sarah Nadi and Xin Xia the Demonstrations Track. Many thanks to Denys Poshyvanyk and Didar Zowghi the Doctoral Symposium Chairs, Federica Sarro, Mohamed Abdelrazek and Yixue Zhao for running the student mentoring programme, Minghui Zhou and Stefano Zacchiroli the artefacts track. Thank you to Andrea Zisman and David Shepherd who ran the Journal First track, Myra Cohen and Hongyu Zhang for running the Technical Briefings track, and Kelly Blincoe and Steve Versteeg for running the Industry Forum. Thank you to Nicole Novielli and Xin Peng for running the posters and informal demonstrations track. Thank you to Betty Cheng and Klaus Pohl who chaired the committee who selected the ICSE Most Influential paper for this year, Muneera Bano and Lingxiao Jiang for running the ACM student research competition, and Xiaoning Du, Mattia Fazzini, Oscar Chaparro and Luís Cruz our SCORE track chairs. Many thanks to Joanna Attlee, Marsha Chechik, Sebastian Uchitel and Nenad Medvidović for organising our ICSE Showcase, and Hoa Dam and Xing Hu for organising the Future of Software Engineering track.

My thanks to Aldeida Aleti and Andrian Marcus who organised the new faculty and researcher symposium, Zhenchang Xing and Bonita Sharif who ran ICSE newcomer events, and to Feifei Chen and Chetan Arora for organising the ICSE networking events. Joshua Garcia and Ita Richardson led development of a rich diversity and inclusion programme, and even evaluated the conference web site for accessibility and inclusivity. Patricia Lago and Marcel Boehme oversaw sustainability initiatives. Kla Tantithamthavorn and Laura Moreno ensured a very vibrant social media presence for the conference over the past few years. Pick Thongtanunam, Matthias Galster and Jonny Low managed a large number of student volunteer applications and due to the hybrid nature of ICSE2023, had to train and supply a large cohort of selected student volunteers. My thanks to Li Li and Niroshine Fernando for overseeing the proceedings process, and to Patrick and Lisa from IEEE CPS for all of their work assembling the ICSE proceedings - with over 1,000 items, no mean feat. Thank you very much to Hourieh Khalajzadeh and Chunyang Chen who worked with Lori, Max and our ICSMA team on the ICSE schedule. Thank you to Anuradha Madugalla and Kevin Ryan our awards chairs, Chun Yong Chong our open science chair, and to Kashumi Madampe our recruitment chair. Finally the team from ICMS Australasia, our Professional Conference Organisers, worked incredibly hard to ensure ICSE was a logistical and organisational success, pre-, during- and post-conference. I would also like to thank Kathy Park from IEEE for all of her help and support.

Thank you to all of the co-located event and workshop organisers for all of your efforts that made the ICSE week programme so much richer. Many thanks to all of the various programme and selection committee members for ICSE tracks, co-located events and workshops who donated their time freely to provide feedback to authors.

I want to gratefully thank our many sponsors and supporters: IEEE Computer Society and IEEE Technical Council on Software Engineering; ACM and ACM Special Interest Group on Software Engineering; the Melbourne Convention Bureau; the State Government of Victoria; the CSIRO; Huawei; Monash University; Meta; Google; AWS; Monash University; Dragon-Testing Technology; IBM; University of Melbourne and RMIT University.

Finally - many thanks to all of the submitters and attendees, without whom there would not be a conference to organise and run.

Enjoy ICSE2023!!

John Grundy
Department of Software Systems and Cybersecurity
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Message from the ICSE 2023 Program Co-Chairs

Welcome to ICSE 2023! It is our great pleasure to introduce the program of the 45th IEEE/ACM International Conference on Software Engineering (ICSE 2023), which will be held in Melbourne, Australia, on May 14-20, 2023.

This year, the technical track received 797 papers, of which 780 entered the reviewing process, and 208 (26.6% acceptance rate) were ultimately accepted, 34 of which after a gatekeeping phase, one was withdrawn after acceptance, making a total of 207 papers in the program.

In the following, we would like to provide some information about the reviewing process.

The role of area chairs and PC members. Since 2021, the ICSE reviewing process has leveraged the help of area chairs. The role of area chairs is to make sure that submissions belonging to a given research area are treated along consistent criteria and standards common to that area. Considering the increasing volume of submissions for some areas, this year we involved ten area chairs: David Lo, Denys Poshyvanyk, Myra Cohen, Corina S. Pasareanu, Abhik Roychoudhury, Walid Maalej, Margaret-Anne Storey, Andrian Marcus, Bram Adams, and Kelly Blincoe.

Area chairs have provided manyfold and significant support to the whole reviewing process. First, they helped us to assign papers to reviewers, and to identify out-of-scope submissions. Then, they supervised the reviewing process for all papers assigned to them, identifying the need for additional expertise, moderating the discussions, and encouraging the convergence toward a decision. Importantly, as also happened for ICSE 2022, there was no single paper on which program co-chairs had to suggest a recommendation because no consensus had been reached. This is particularly important because it means that the decision was always taken without having knowledge about the paper's authorship (known to program chairs only). Overall, each area chair had a workload comparable to those of program co-chairs of medium-sized conferences. They all did a terrific job, and we are extremely grateful to them!

The ICSE 2023 program committee was composed of 194 PC members, with a wide diversity in terms of expertise, geographic distribution, and gender. They did an incredible job in providing timely and constructive reviewers in a relatively tight period of time (about 6 weeks), and in participating in the online discussion. Overall, PC members have written a total of 2,347 reviews and posted 10,083 comments.

Among the PC members, based on the inputs from area chairs and other co-reviewers, we recognize the following 23 as distinguished reviewers: Amjed Tahir, Andreas Vogelsang, Andrzej Wasowski, Aurdris Mockus, Bogdan Vasilescu, Brittany Johnson, Caitlin Sadowski, Christine Julien, Diomidis Spinellis, Igor Steinmacher, Jennifer Horkoff, Julia Lawall, Kevin Moran, Kla Tantithamthavorn, Klas Stool, Leandro Minku, Leonardo Mariani, Marcel Boehme, Philipp Leitner, Rachel Tzoref-Brill, Sebastian Baltes, Sebastian Proksch, and Xavier Franch.

Reviewing process. After a first scrutiny from us program co-chairs (to check compliance with the format, anonymization, and scope), after a plagiarism analysis, and after a further scope check from the area chairs when needed, each manuscript received at least three reviews. After a first

preliminary discussion (which sometimes led to the addition of additional reviews), the authors of all papers had six days to read the reviews and provide a response to the reviewers. Reviews were then thoroughly discussed, taking into account the authors' responses, where a response was provided. For a small subset of papers, where further questions emerged during the discussion or further reviews were required, a second response period was then opened.

After a consensus was reached, for each paper, one of the reviewers was also asked to write a metareview, explaining the reasons for acceptance or rejection, also in relation to what has been discussed behind the scenes. Some papers (35) were asked to undergo a gatekeeping process after the notification, i.e., a revised version was re-reviewed by a shepherd. Of those, 34 were accepted.

Reviewing guidelines. Following what was done in the previous ICSE, we provided PC members (as well as authors) with a detailed set of guidelines, with the aim of improving the consistency in the provided reviews, setting reviewing criteria along different dimensions (technical soundness, novelty, significance, verifiability and transparency, and presentation), and avoiding, wherever possible, known antipatterns. We are so grateful to the authors of the initial sets of guidelines (Amy Ko and Jan Vahrenhold program co-chairs of ICER 2021, Andreas Zeller and Daniela Damian program co-chairs of ICSE 2022) which we have evolved and adapted to our own reviewing process. We hope that this initiative continues over future ICSE conferences (it is already happening, at the time of this writing, for ICSE 2024).

Open access policy and artifacts check. Continuing the tradition of previous ICSE conferences, we adopted the open-access policy. Specifically, each submitted manuscript was supposed to provide the availability of data (and/or other artifacts aimed at supporting the research reproducibility). Where this was not possible, the authors were expected to explain the reasons. For the first time, this year, we asked one of the reviewers of each paper to scrutinize the provided artifacts and check whether they are in line with what was declared at submission time. In our opinion, this helped to encourage the submission of complete artifacts, and are happy that forthcoming conferences are also adopting this initiative.

Distinguished papers. It is a tradition of SIGSOFT-sponsored conferences to award less than 10% of the accepted papers with a Distinguished Paper Award. This year, we have selected 10 outstanding papers:

1. Do I Belong? Modeling Sense of Virtual Community Among Linux Kernel Contributors. *Bianca Trinkenreich, Klaas-Jan Stol, Anita Sarma, Daniel M German, Marco Gerosa, and Igor Steinmacher*
2. Compatible Remediation on Vulnerabilities from Third-Party Libraries for Java Projects. *Lyuye Zhang, Chengwei Liu, Zhengzi Xu, Sen Chen, Lingling Fan, Lida Zhao, Jiahui Wu, and Yang Liu*
3. A Qualitative Study on the Implementation Design Decisions of Developers. *Jenny T. Liang, Maryam Arab, Minhyuk Ko, Amy J. Ko, and Thomas D. LaToza*
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Lori Pollock,
University of Delaware, DE, USA

Massimiliano Di Penta,
University of Sannio, Italy

Message from the ICSE 2023 Program Co-Chairs

Welcome to ICSE 2023! It is our great pleasure to introduce the program of the 45th IEEE/ACM International Conference on Software Engineering (ICSE 2023), which will be held in Melbourne, Australia, on May 14-20, 2023.

This year, the technical track received 797 papers, of which 780 entered the reviewing process, and 208 (26.6% acceptance rate) were ultimately accepted, 34 of which after a gatekeeping phase, one was withdrawn after acceptance, making a total of 207 papers in the program.

In the following, we would like to provide some information about the reviewing process.

The role of area chairs and PC members. Since 2021, the ICSE reviewing process has leveraged the help of area chairs. The role of area chairs is to make sure that submissions belonging to a given research area are treated along consistent criteria and standards common to that area. Considering the increasing volume of submissions for some areas, this year we involved ten area chairs: David Lo, Denys Poshyvanyk, Myra Cohen, Corina S. Pasareanu, Abhik Roychoudhury, Walid Maalej, Margaret-Anne Storey, Andrian Marcus, Bram Adams, and Kelly Blincoe.

Area chairs have provided manyfold and significant support to the whole reviewing process. First, they helped us to assign papers to reviewers, and to identify out-of-scope submissions. Then, they supervised the reviewing process for all papers assigned to them, identifying the need for additional expertise, moderating the discussions, and encouraging the convergence toward a decision. Importantly, as also happened for ICSE 2022, there was no single paper on which program co-chairs had to suggest a recommendation because no consensus had been reached. This is particularly important because it means that the decision was always taken without having knowledge about the paper's authorship (known to program chairs only). Overall, each area chair had a workload comparable to those of program co-chairs of medium-sized conferences. They all did a terrific job, and we are extremely grateful to them!

The ICSE 2023 program committee was composed of 194 PC members, with a wide diversity in terms of expertise, geographic distribution, and gender. They did an incredible job in providing timely and constructive reviewers in a relatively tight period of time (about 6 weeks), and in participating in the online discussion. Overall, PC members have written a total of 2,347 reviews and posted 10,083 comments.

Among the PC members, based on the inputs from area chairs and other co-reviewers, we recognize the following 23 as distinguished reviewers: Amjed Tahir, Andreas Vogelsang, Andrzej Wasowski, Aurdris Mockus, Bogdan Vasilescu, Brittany Johnson, Caitlin Sadowski, Christine Julien, Diomidis Spinellis, Igor Steinmacher, Jennifer Horkoff, Julia Lawall, Kevin Moran, Kla Tantithamthavorn, Klas Stool, Leandro Minku, Leonardo Mariani, Marcel Boehme, Philipp Leitner, Rachel Tzoref-Brill, Sebastian Baltes, Sebastian Proksch, and Xavier Franch.

Reviewing process. After a first scrutiny from us program co-chairs (to check compliance with the format, anonymization, and scope), after a plagiarism analysis, and after a further scope check from the area chairs when needed, each manuscript received at least three reviews. After a first

preliminary discussion (which sometimes led to the addition of additional reviews), the authors of all papers had six days to read the reviews and provide a response to the reviewers. Reviews were then thoroughly discussed, taking into account the authors' responses, where a response was provided. For a small subset of papers, where further questions emerged during the discussion or further reviews were required, a second response period was then opened.

After a consensus was reached, for each paper, one of the reviewers was also asked to write a metareview, explaining the reasons for acceptance or rejection, also in relation to what has been discussed behind the scenes. Some papers (35) were asked to undergo a gatekeeping process after the notification, i.e., a revised version was re-reviewed by a shepherd. Of those, 34 were accepted.

Reviewing guidelines. Following what was done in the previous ICSE, we provided PC members (as well as authors) with a detailed set of guidelines, with the aim of improving the consistency in the provided reviews, setting reviewing criteria along different dimensions (technical soundness, novelty, significance, verifiability and transparency, and presentation), and avoiding, wherever possible, known antipatterns. We are so grateful to the authors of the initial sets of guidelines (Amy Ko and Jan Vahrenhold program co-chairs of ICER 2021, Andreas Zeller and Daniela Damian program co-chairs of ICSE 2022) which we have evolved and adapted to our own reviewing process. We hope that this initiative continues over future ICSE conferences (it is already happening, at the time of this writing, for ICSE 2024).

Open access policy and artifacts check. Continuing the tradition of previous ICSE conferences, we adopted the open-access policy. Specifically, each submitted manuscript was supposed to provide the availability of data (and/or other artifacts aimed at supporting the research reproducibility). Where this was not possible, the authors were expected to explain the reasons. For the first time, this year, we asked one of the reviewers of each paper to scrutinize the provided artifacts and check whether they are in line with what was declared at submission time. In our opinion, this helped to encourage the submission of complete artifacts, and are happy that forthcoming conferences are also adopting this initiative.

Distinguished papers. It is a tradition of SIGSOFT-sponsored conferences to award less than 10% of the accepted papers with a Distinguished Paper Award. This year, we have selected 10 outstanding papers:

1. Do I Belong? Modeling Sense of Virtual Community Among Linux Kernel Contributors. *Bianca Trinkenreich, Klaas-Jan Stol, Anita Sarma, Daniel M German, Marco Gerosa, and Igor Steinmacher*
2. Compatible Remediation on Vulnerabilities from Third-Party Libraries for Java Projects. *Lyuye Zhang, Chengwei Liu, Zhengzi Xu, Sen Chen, Lingling Fan, Lida Zhao, Jiahui Wu, and Yang Liu*
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- David Lo, *Singapore Management University, Singapore*
- Denys Poshyvanyk, *College of William and Mary, United States.*

Area Chairs for Analysis and Testing:

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